Aariz Khan

Artificial Intelligence & Machine Learning Engineer

B.Tech, Artificial Intelligence & Machine Learning — CMR University, Bengaluru

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SUMMARY

Artificial intelligence and machine learning engineer who designs and ships complete AI systems. Three live web applications — retrieval-augmented chat, automated report analysis and semantic resume matching — alongside a voice-controlled desktop agent covered by 1,932 tests. Works across the full path from data preparation and model evaluation to REST APIs and deployment.

TECHNICAL SKILLS

AI & Machine Learning	PyTorch, scikit-learn, regression (linear, logistic), classification (KNN, SVM), feature engineering, model evaluation (ROC-AUC, confusion matrix, precision / recall / F1, R-squared)
NLP & Retrieval	Retrieval-augmented generation, embeddings, semantic search, Sentence-Transformers, vector search (pgvector), hybrid retrieval and reranking, speech recognition (Whisper)
Data Science & Platforms	pandas, NumPy, exploratory data analysis, data cleaning, Matplotlib, seaborn, Plotly, Linux, Kubernetes, REST APIs, PostgreSQL, MongoDB, SQLite, Supabase
Languages & Tools	Python, TypeScript, SQL, Git, GitHub, Jupyter, Google Colab, VS Code, pytest, FastAPI, Streamlit, PySide6 (Qt), Next.js, React

PROJECTS

AARIZ AI — Retrieval-Augmented Document Chatbot LIVE

Next.js · TypeScript · PostgreSQL + pgvector · Supabase · Ollama · Gemini · aariz-ai-three.vercel.app · github.com/Aariz-a-z/aariz-ai

- Answers questions from your own documents and shows which sources it used.
- Reads 10 file types including scanned PDFs, splits them into chunks, and stores them as embeddings in PostgreSQL with pgvector.
- Searches by meaning and keyword together, then reranks. Found the right passage 96.2% of the time across 14 test runs.

ProjectRevive AI — Report Analysis & Recommendations LIVE

Python · FastAPI · Next.js · TypeScript · projectrevive-ai.vercel.app · github.com/Aariz-a-z/projectrevive-ai

- Reads an AI/ML project report and finds what is weak in it, tracing every finding back to the page it came from.
- Keeps what the report says separate from what the system inferred, dropping any finding whose quote cannot be verified.
- Combines 14 deterministic rules with a model to rank fixes from a catalogue of 22 improvements, behind Argon2id hashing and two layers of rate limiting.

AI Resume Analyzer LIVE

Python · Sentence-Transformers · Streamlit · PyMuPDF · aariz-ai-resume-analyzer.streamlit.app · github.com/Aariz-a-z/AI-RESUME-ANALYZER

- Compares a resume with a job description and scores how well they match, using sentence embeddings rather than keyword matching.

AARMA — Voice-Controlled Desktop AI Agent

Python · PySide6 (Qt 6) · SQLite · faster-whisper · edge-tts · pytest · github.com/Aariz-a-z/aarma-desktop-agent

- A Windows assistant that takes typed or spoken commands and runs 103 actions.
- Every action has its own permission setting and a risk level. Risky ones ask before they run.
- Speech recognition runs on the computer, so audio never leaves the machine. Covered by 1,932 tests.

Other projects **Sales Forecasting** · **Wine Quality Classification** · **Travel Planner** · **Python Fundamentals**

EXPERIENCE

AI Trainer · Handshake AI — Remote — Jul 2026 - Sep 2026

- Created and evaluated software-engineering benchmark tasks for AI coding agents — authoring each problem, its reference solution and the tests that verify it, then running evaluations and reviewing every task for difficulty, fairness and correctness, delivered through GitHub pull-request workflows.

EDUCATION

B.Tech, Artificial Intelligence & Machine Learning · CMR University, Bengaluru — 2023 - 2027 · **CGPA 8.8**

Pre-University (MPC) · Sri Chaitanya College, Kadapa — 2021 - 2023 · **70%**

Secondary School · Keshava Reddy School, Nandyal — 2021 · **CGPA 9.8**

Languages English, Hindi, Urdu, Telugu